

Touchscreen And GLCD-Based HOME AUTOMATION

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This project controls home electrical appliances using a touchscreen input device. The system comprises two relays for controlling a bulb and a fan as an example. The touchscreen placed on a graphical LCD acts as a control panel for sending control commands through a pair of wireless radio frequency (RF) communication modules. The touchscreen panel on the transmitter side interfaced to a microcontroller sends on/off commands to the receiver where the loads/appliances are connected. By touching the specified portion on the touchscreen panel, the loads are turned on/off remotely through the wireless RF modules.

Circuit and working

The system is divided into transmitter and receiver sections.

Transmitter section. Block diagram of the transmitter section is shown in Fig. 1. The power supply unit is used to power the microcontroller, RF Tx module and graphics LCD (GLCD). The GLCD is interfaced with the microcontroller. The touchscreen is transparent and is hence placed on top of the GLCD. Data obtained from the touchscreen is processed by the microcontroller to generate a 4-digit binary code. This code is encoded by the encoder and transmitted through the RF transmitter module.

Circuit diagram of the transmitter section is shown in Fig. 2. It is built around transformer X1, bridge rectifier BR1, 128x64 GLCD (LCD1), 4-wire touchscreen (TS1), 5V voltage regulator 7805

(IC1), ATmega16A microcontroller (IC2), encoder HT12E (IC3), 433MHz RF transmitter module (TX1), an antenna (ANT.1) and a few other components.

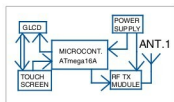
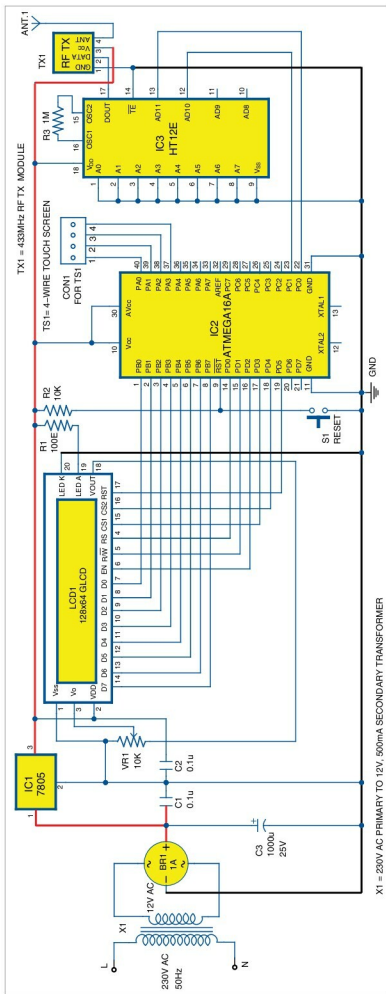


Fig. 1: Block diagram of the transmitter



X1 = 230V AC PRIMARY TO 12V 500mA SECONDARY TRANSFORMER

Fig. 2: Circuit diagram of the transmitter